PCA and Dimensionality Reduction

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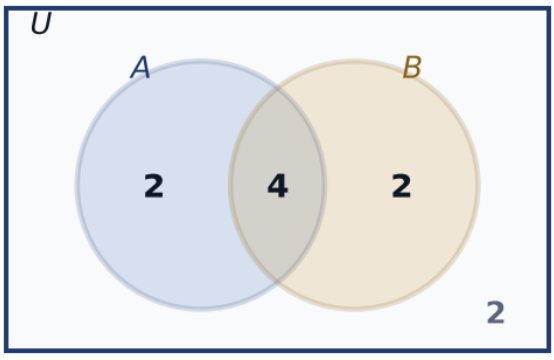
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider a dataset with n data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, where each $\mathbf{x}_i \in \mathbb{R}^p$. The sample covariance matrix is given by $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$, where $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i$ is the sample mean. Suppose PCA is performed on this dataset, resulting in principal components defined by the eigenvectors of $\mathbf{S}$ and corresponding variances given by the eigenvalues. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be the orthonormal eigenvectors of $\mathbf{S}$ with associated eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. The first principal component captures the direction of maximum variance, λ_1, and the second principal component captures the next largest variance, λ_2, orthogonal to the first. If we decide to reduce the dimensionality from p to k by keeping only the first k principal components, the proportion of the total variance retained is given by $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i}$. Suppose for a particular dataset, the eigenvalues of the covariance matrix are $\{10.5, 5.2, 2.1, 0.8, 0.4\}$. If we aim to retain at least 90% of the total variance, what is the minimum number of principal components required? Show your complete algebraic work.

2 [Foundational] • Target: Data Centering and the Covariance Matrix

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by: $S = (4, -2; -2, 1)$ Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = (a; b)$. The variance explained by this principal component is given by the corresponding eigenvalue λ_1. If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$? Show your complete algebraic work.



3 [Foundational] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

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4 [Foundational] • Target: Explained Variance and Scree Analysis

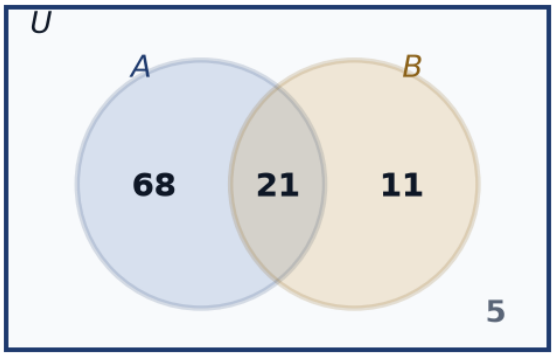
A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 64$, $\lambda_2 = 23$, and $\lambda_3 = 13$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

5 [Foundational] • Target: SVD Connection and Applications

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 66$, $\lambda_2 = 22$, and $\lambda_3 = 12$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

6 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.



7 [Foundational] • Target: Data Centering and the Covariance Matrix

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 70$, $\lambda_2 = 20$, and $\lambda_3 = 10$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 72$, $\lambda_2 = 19$, and $\lambda_3 = 9$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

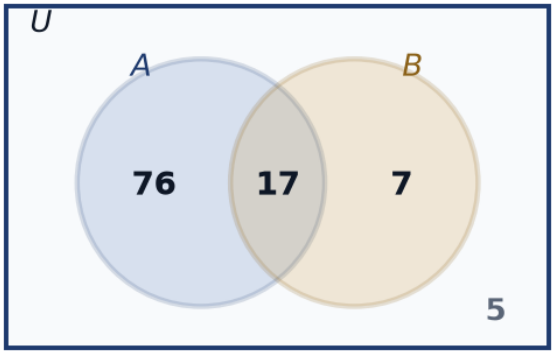
9 [Intermediate] • Target: Explained Variance and Scree Analysis

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 74$, $\lambda_2 = 18$, and $\lambda_3 = 8$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

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10 **[Intermediate]** • **Target: SVD Connection and Applications**

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 76$, $\lambda_2 = 17$, and $\lambda_3 = 7$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.



11 **[Intermediate]** • **Target: PCA & Dimensionality Reduction Practice Exercises**

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by: $\Sigma = (10, 5; 5, 20)$ What is the variance of the first feature, X_1? Show your complete algebraic work.

12 **[Intermediate]** • **Target: Data Centering and the Covariance Matrix**

A centered feature matrix $\mathbf{X}$ has shape $N \times 14$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

13 **[Intermediate]** • **Target: Eigenvalues, Eigenvectors and Variance Maximization**

A centered feature matrix $\mathbf{X}$ has shape $N \times 18$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

14 **[Intermediate]** • **Target: Explained Variance and Scree Analysis**

A centered feature matrix $\mathbf{X}$ has shape $N \times 22$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

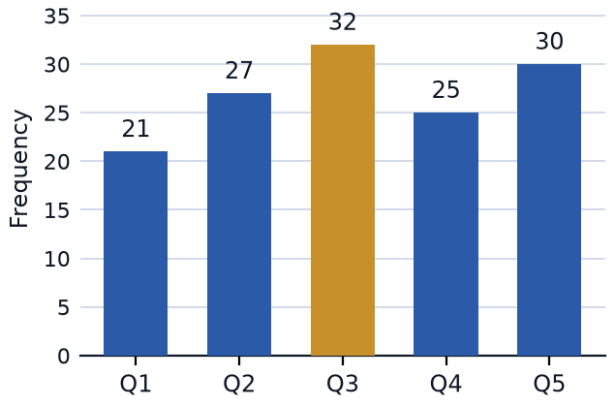
15 **[Challenging]** • **Target: SVD Connection and Applications** • **Cross-Lesson: 1.01 Vectors and Embeddings in AI**

A centered feature matrix $\mathbf{X}$ has shape $N \times 26$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

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16 **[Challenging]** • **Target: PCA & Dimensionality Reduction Practice Exercises** • **Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

A centered feature matrix $\mathbf{X}$ has shape $N \times 30$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.



17 **[Challenging]** • **Target: Data Centering and the Covariance Matrix** • **Cross-Lesson: 3.01 Gradient Descent and Optimization**

A centered feature matrix $\mathbf{X}$ has shape $N \times 34$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

18 **[Challenging]** • **Target: Eigenvalues, Eigenvectors and Variance Maximization** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

A centered feature matrix $\mathbf{X}$ has shape $N \times 38$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

19 **[Challenging]** • **Target: Explained Variance and Scree Analysis** • **Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

A centered feature matrix $\mathbf{X}$ has shape $N \times 42$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

20 **[Challenging]** • **Target: SVD Connection and Applications** • **Cross-Lesson: 6.01 Linear Regression and OLS**

Consider a 2D data point $\mathbf{v} = [3;\ -4]$. What is the squared Euclidean norm (or squared magnitude) of this vector? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{30} \) onto its top \(2 \) principal components. What is the dimension of each transformed data point?

a. \(15 \)

b. \(30 \)

c. \(28 \)

d. \(2 \)

22 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{40} \) onto its top \(4 \) principal components. What is the dimension of each transformed data point?

a. \(20 \)

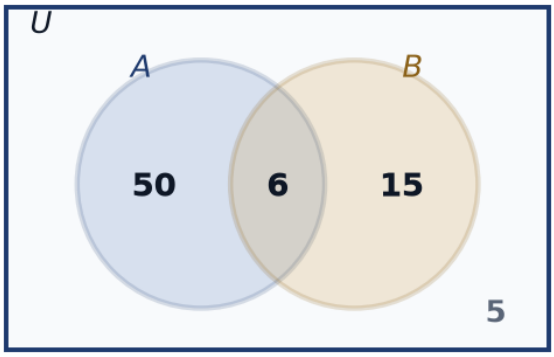
b. \(40 \)

c. \(36 \)

d. \(4 \)

23 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{50} \) onto its top \(6 \) principal components. What is the dimension of each transformed data point?



a. \(25 \)

b. \(50 \)

c. \(44 \)

d. \(6 \)

24 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{60} \) onto its top \(8 \) principal components. What is the dimension of each transformed data point?

a. \(30 \)

b. \(60 \)

c. \(52 \)

d. \(8 \)

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25 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{70}$ onto its top 10 principal components. What is the dimension of each transformed data point?

a. 10

b. 70

c. 60

d. 35

26 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{80}$ onto its top 12 principal components. What is the dimension of each transformed data point?

a. 12

b. 80

c. 68

d. 40

27 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{90}$ onto its top 14 principal components. What is the dimension of each transformed data point?

a. 14

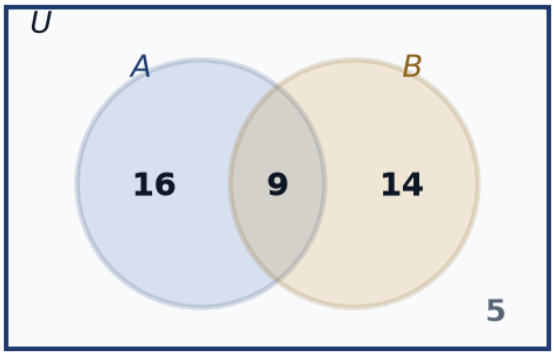
b. 90

c. 76

d. 45

28 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{100}$ onto its top 16 principal components. What is the dimension of each transformed data point?



a. 100

b. 16

c. 84

d. 50

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Data Centering and the Covariance Matrix**

Why must data features be centered (subtracting feature means) prior to performing Principal Component Analysis?

a. So the first principal component aligns with the axis of maximum variance rather than the offset from origin

b. To make the data non-linear

c. To ensure all eigenvalues are equal to 1

d. To reduce the matrix dimension

30 **[Intermediate]** • **Target: Data Centering and the Covariance Matrix**

For a zero-mean data matrix $\mathbf{X} \in \mathbf{R}^{N \times D}$, state the sample covariance matrix $\mathbf{\Sigma}$.

a. $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$

b. $\mathbf{\Sigma} = \frac{1}{D-1}\mathbf{X}\mathbf{X}^T$

c. $\mathbf{\Sigma} = \mathbf{X}^T\mathbf{X}$

d. $\mathbf{\Sigma} = \frac{1}{N}\mathbf{X}^2$

31 **[Intermediate]** • **Target: Data Centering and the Covariance Matrix**

What is the mathematical symmetry property of the covariance matrix $\mathbf{\Sigma}$?

a. It is symmetric and positive semi-definite (all eigenvalues $\lambda_i \geq 0$)

b. It is skew-symmetric

c. It is an orthogonal matrix

d. It has negative trace

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32 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A 2×2 covariance matrix has entries $\Sigma_{11} = 4$, $\Sigma_{22} = 1$, and covariance $\Sigma_{12} = \Sigma_{21} = 2$. Find its eigenvalues.

a. $\lambda_1 = 5, \quad \lambda_2 = 0$

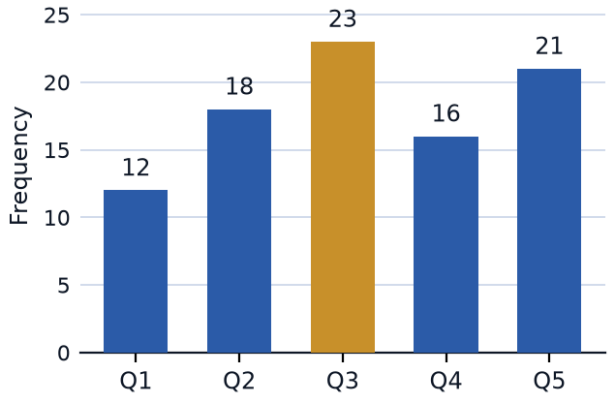
b. $\lambda_1 = 4, \quad \lambda_2 = 1$

c. $\lambda_1 = 3, \quad \lambda_2 = 2$

d. $\lambda_1 = 2, \quad \lambda_2 = 2$

33 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

What is the geometric meaning of the first eigenvector $\mathbf{v}_1$ corresponding to largest eigenvalue λ_1?



a. The direction along which the data exhibits maximum variance

b. The direction perpendicular to all data points

c. The direction with the lowest information content

d. The vector of class centroids

34 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

If $\mathbf{v}$ is an eigenvector of $\mathbf{\Sigma}$ with eigenvalue λ, what is the variance of data projected onto $\mathbf{v}$?

a. λ

b. $\sqrt{\lambda}$

c. λ^2

d. 1

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35 **[Intermediate]** • **Target: Explained Variance and Scree Analysis**

A dataset has eigenvalues $\lambda_1 = 70, \lambda_2 = 20, \lambda_3 = 10$. What percentage of total variance is explained by the first two principal components?

a. 90%

b. 70%

c. 80%

d. 20%

36 **[Intermediate]** • **Target: Explained Variance and Scree Analysis**

How is an 'elbow' in a scree plot (eigenvalues plotted against component index) used in practice?

a. To select the optimal cutoff k where adding further components yields diminishing returns

b. To detect outliers in the target labels

c. To determine the learning rate for gradient descent

d. To test for multicollinearity

37 **[Intermediate]** • **Target: Explained Variance and Scree Analysis**

If a dataset has 100 features and you retain the top 10 principal components, by what percentage is feature dimensionality reduced?

a. 90%

b. 10%

c. 50%

d. 80%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Given the Singular Value Decomposition $\mathbf{X} = \mathbf{USV}^T$, how are the principal component directions $\mathbf{v}_i$ directly obtained?

a. As the right singular vectors (columns of $\mathbf{V}$)

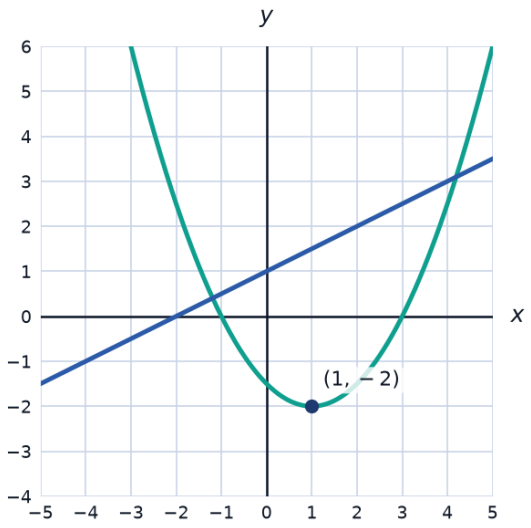
b. As the left singular vectors (columns of $\mathbf{U}$)

c. As the diagonal elements of $\mathbf{S}$

d. As the eigenvalues of $\mathbf{U}$

39 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

What is the mathematical relation between singular values s_i of $\mathbf{X}$ and eigenvalues λ_i of covariance matrix $\mathbf{\Sigma}$?



a. $\lambda_i = \frac{s_i^2}{N-1}$

b. $\lambda_i = s_i$

c. $\lambda_i = \sqrt{s_i}$

d. $\lambda_i = s_i(N-1)$

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40 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 6.01 Linear Regression and OLS

Name a major practical application of PCA in image processing and computer vision.

a. Eigenfaces: facial compression and feature representation in lower-dimensional eigenspace

b. Image sharpening through edge convolution

c. Color palette inversion

d. Bilinear image interpolation

41 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{50}$ onto its top 6 principal components. What is the dimension of each transformed data point?

a. 25

b. 50

c. 44

d. 6

42 [Challenging] • Target: Data Centering and the Covariance Matrix • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider a linear transformation represented by the matrix $A = (3, 0; 0, -2)$. If $\mathbf{v} = (1; 1)$ is an eigenvector of A, what is the corresponding eigenvalue?

a. 3

b. -2

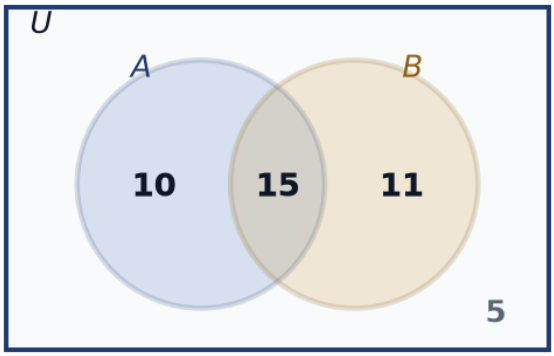
c. 1

d. 5

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43 [Challenging] • Target: Eigenvalues, Eigenvectors and Variance Maximization • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider a dataset of n data points, where each data point is a vector in $\mathbb{R}^D$. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space $\mathbb{R}^d$. The mapping function for a single data point $\mathbf{x} \in \mathbb{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathbb{R}^D \to \mathbb{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathbb{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.



a. 15

b. 10

c. 1

d. 25

44 [Challenging] • Target: Explained Variance and Scree Analysis • Cross-Lesson: 3.01 Gradient Descent and Optimization

The vector $\mathbf{v} = (-2; 1)$ is an eigenvector of the matrix $A = (3, 2; 1, 4)$. What is the corresponding eigenvalue λ?

a. 2

b. 5

c. -2

d. 7

45 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = (1; 2)$. If the trace of A is 7, find the other eigenvalue of A.

a. 1

b. 4

c. 10

d. 21

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